

Submission Worksheet

CLICK TO GRADE

<https://learn.ethereallab.app/assignment/IT114-005-F2024/it114-module-2-java-problems/grade/bna24>

Course: IT114-005-F2024

Assignment: [IT114] Module 2 Java Problems

Student: Bakr A. (bna24)

Submissions:

Submission Selection

1 Submission [submitted] 9/23/2024 7:49:31 PM

Instructions

^ COLLAPSE ^

Overview Video: <https://youtu.be/4M8Di5jrcZQ>

Guide:

1. Make sure you're in the main branch locally and `git pull origin main` any pending changes.
2. Make a new branch per the recommended branch name below (`git checkout -b ...`).
3. Create a folder in your local repo called `Module2`
4. Grab the template code from <https://gist.github.com/MattToegel/fdd2b37fa79a06ace9dd259ac82728b6>.
5. Create individual Java files for each problem and save the files inside the `Module2` folder.
 1. They should end with the file extension in lowercase `.java`.
6. Move the unedited template files to GitHub.
 1. `git add .`
 2. `git commit -m "adding template files"`
 3. `git push origin branch_name` (see below).
 4. Create and open a pull request from the homework branch to main (leave it open until later steps).
7. Note: As you work, it's recommended to add/commit at least after each solution is done (i.e., 3+ times in this case).
 1. Make sure the files are saved before doing this.
 2. A file is unsaved if you see a white dot in the tab where the filename shows in VS Code
8. Fill in the items in the worksheet below (save as often as necessary).
9. Once finished, export the worksheet.
10. Add the output file to any location of your choice in your repository folder (i.e., a `Module2` folder).
11. Check that git sees it via `git status`.

12. If everything is good, continue to submit.
 1. Track the file(s) via `git add`.
 2. Commit the changes via `git commit` (don't forget the commit message).
 3. Push the changes to GitHub via `git push` (don't forget to refer to the proper branch).
 4. Create a pull request from the homework related branch to main (i.e., `main <- "homework branch"`).
 5. Open and complete the merge of the pull request (it should turn purple).
 6. Locally checkout main and pull the latest changes (to prepare for future work).
13. Take the same output file and upload it to Canvas.

Branch name: M2-Java-Problems

Group



Group: Problem 1
Tasks: 1
Points: 3

^ COLLAPSE ^

Task



Group: Problem 1
Task #1: Screenshot of the Problem 1 Solved Code and Output
Weight: ~100%
Points: ~3.00

^ COLLAPSE ^

Details:

Only make edits where the template code mentions.

Solution should ensure that any passed in array will have only the odd values output.
Requires at least 2 screenshots (code + output from terminal)



Columns: 1

Sub-Task



Group: Problem 1
Task #1: Screenshot of the Problem 1 Solved Code and Output
Sub Task #1: Screenshot the output of the solved problem

Task Screenshots

Gallery Style: 2 Columns

4

2

1

```
task@ubuntu: ~/M2Java64 -/tns24-TT114-005 (M2-Java-Problems)
$ java M2.Problem
Processing Array:[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

```

Odds output:
1 3 5 7 9
End process
Processing Array:[0, 1, 3, 5, 7, 9, 2, 4, 6, 8, 10]
Odds output:
1 3 5 7 9
End process
Processing Array:[10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0]
Odds output:
9 7 5 3 1
End process
Processing Array:[0, 0, 1, 1, 2, 2, 3, 3, 4, 4, 5, 5, 6, 6, 7, 7, 8, 8, 9, 9, 10, 10]
Odds output:
1 1 3 3 5 5 7 7 9 9
End process

```

Problem 1 Terminal Output

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

Sub-Task

Group: Problem 1

100%

Task #1: Screenshot of the Problem 1 Solved Code and Output

Sub Task #2: Screenshot the code solution (ucid/date must be included as a comment)

Task Screenshots

Gallery Style: 2 Columns

4

2

1

```

// 9/23/2024
//UCID: bna24
for (int i = 0; i < arr.length; i++) {
    if (arr[i] % 2 == 1) {
        System.out.print(arr[i] + " ");
    }
} <- #27-31 for (int i = 0; i < arr.length; i++)

```

Problem 1 Code

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

Task Response Prompt

Explain in concise steps how this logically works

Response:

Logically this works because it first, creates a for loop that is the length of the array given in order to access each value. Then it goes to an If statement which passes each value through a condition, to see if it is an odd value (if the remainder is 1 it means it is odd), then prints out the value.

End of Task 1

End of Group: Problem 1

Task Status: 1/1

Group

Group: Problem 2

Tasks: 1

Points: 3

100%

^ COLLAPSE ^

Task



Group: Problem 2

Task #1: Screenshot of the Problem 2 Solved Code and Output

Weight: ~100%

Points: ~3.00

^ COLLAPSE ^

Details:

Only make edits where the template code mentions.

Solution should ensure that any passed in array will have its values summed AND the final result converted to two decimal places (i.e., 0.10, 1.00, 1.01).



Columns: 1

Sub-Task



Group: Problem 2

Task #1: Screenshot of the Problem 2 Solved Code and Output

Sub Task #1: Screenshot the output of the solved problem

Task Screenshots

Gallery Style: 2 Columns

4 2 1

```
Processing Array:[10.001, 11.501, 0.011, 5.991, 16.121, 0.131, 100.981, 1.001]
Total is 145.83
End process
Processing Array:[1.99, 1.99, 0.99, 1.99, 0.99, 1.99, 0.99, 0.99]
Total is 11.92
End process
Processing Array:[0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01]
Total is 0.1
End process
Processing Array:[10.01, -12.22, 0.25, 19.2, -5.15, 3.12]
Total is 15.21
End process
```

Code Output of problem 2

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

Sub-Task



Group: Problem 2

Task #1: Screenshot of the Problem 2 Solved Code and Output

Sub Task #2: Screenshot the code solution (ucid/date must be included as a comment)

Task Screenshots

Gallery Style: 2 Columns

4 2 1

```
// 9/23/2024
// UCID: bna24
for (int i = 0; i < arr.length; i++){
    total += arr[i];
}

// multiply by 100 to move decimal points so that math.round can work
// Then divide to move the decimal back
total = Math.round(total * 100.0) / 100.0;

// ensure rounding is to two decimal places (i.e., 0.10, 0.01, 1.00)
// set the double to a string variable while formatting 2 decimal places
totalOutput = String.format(format: "%.2f", total);
```

Output of Code

Caption(s) (required) ✓

Caption Hint: *Describe/highlight what's being shown*

Task Response Prompt

Explain in concise steps how this logically works

Response:

First there is a for loops that itirates through each array and adds each element of the array to total. Then total is passed through math.round which (when total multiplied by 100 moves the decimals points) it rounds to the nearest int and divides it so that the decimal points are back where they were. Lastly, the totalOutput variable is formatted through the .format command which goes through %.2f where the 2 means that you want to display 2 decimal places and f is for floating point number.

End of Task 1

End of Group: Problem 2

Task Status: 1/1

Group

100%

Group: Problem 3

Tasks: 1

Points: 3

^ COLLAPSE ^

Task

100%

Group: Problem 3

Task #1: Screenshot of the Problem 3 Solved Code and Output

Weight: ~100%

Points: ~3.00

^ COLLAPSE ^

Details:

Only make edits where the template code mentions.

Solution should ensure that any passed in array will have its values converted to a positive version of the value AND converted back to the original data type.

Sub-Task

Group: Problem 3

100%

Task #1: Screenshot of the Problem 3 Solved Code and Output

Sub Task #1: Screenshot the output of the solved problem

Task Screenshots

Gallery Style: 2 Columns

4 2 1

```

bakra@bakra MINKA64 ~/bna24-11114-005 (P2-Java-Problems)
$ java M2.Problem1
Processing Array: [-1, -2, -3, -4, -5, -6, -7, -8, -9, -10]
Result: 1 (1), 2 (1), 3 (1), 4 (1), 5 (1), 6 (1), 7 (1), 8 (1), 9 (1), 10 (1)
Processing Array: [-1, 1, -2, 2, 3, -3, -4, 5]
Result: 1 (1), 1 (1), 2 (1), 2 (1), 3 (1), 3 (1), 4 (1), 5 (1)
Processing Array: [-0.01, -1.0E-4, -0.15]
Result: 0.01 (D), 1.0E-4 (D), 0.15 (D)
Processing Array: [-1, 2, -3, 4, -5, 5, -6, 6, -7, 7]
Result: 1 (S), 2 (S), 3 (S), 4 (S), 5 (S), 5 (S), 6 (S), 6 (S), 7 (S), 7 (S)

```

Output of Problem 3

Caption(s) (required) ✓

Caption Hint: Describe/highlight what's being shown

Sub-Task

Group: Problem 3

100%

Task #1: Screenshot of the Problem 3 Solved Code and Output

Sub Task #2: Screenshot the code solution (ucid/date must be included as a comment)

Task Screenshots

Gallery Style: 2 Columns

4 2 1

```

// 9/25/2024
// UCID: bna24
for (int i = 0; i < arr.length; i++) {
    if (arr[i] instanceof Integer) {
        output[i] = Math.abs((Integer) arr[i]);
    } else if (arr[i] instanceof Double) {
        output[i] = Math.abs((Double) arr[i]);
    } else if (arr[i] instanceof String) {
        output[i] = String.valueOf(Math.abs(Integer.parseInt((String) arr[i])));
    } else {
        output[i] = arr[i];
    }
}
} < #34-44 for (int i = 0; i < arr.length; i++)

```

Problem 3 Code

Caption(s) (required) ✓

Caption Hint: Describe/highlight what's being shown

Task Response Prompt

Explain in concise steps how this logically works

Response:

It iterates over each element in the input array, checking its type. If the element is an Integer or Double, it converts it to a positive value using Math.abs(). If it's a String, it converts the string to an integer, makes it positive, and converts

it back to a string. Finally, the method prints the output array, formatted with the values and their data types.

End of Task 1

End of Group: Problem 3

Task Status: 1/1

Group



Group: Reflection

Tasks: 3

Points: 1

^ COLLAPSE ^

Task



Group: Reflection

Task #1: Reflect on your experience

Weight: ~33%

Points: ~0.33

^ COLLAPSE ^

i Details:

Talk about any issues you had, how you resolved them, and anything you learned during this process.

Provide concrete details/examples. At least a few sentences.



≡ Task Response Prompt

Response:

For problem 2, I had issues figuring out how Math.round works because I havent used it in so long. I had to look up the documentation and an example of how it worked to see how to use it. For problem 3, I needed to find a way to compare any data type against another data type. I rarely used "instanceof" in CS113, and thats true for a bunch of other methods/commands but I am learning how to use them in this class.

End of Task 1

Task



Group: Reflection

Task #2: Include the pull request link for this branch

Weight: ~33%

Points: ~0.33

^ COLLAPSE ^

i Details:

Details.

The correct link will end with /pull/ and a number.



Task URLs

URL #1

<https://github.com/BakrAbushaar/bna24-IT114-005/pull/2>

URL

<https://github.com/BakrAbushaar/bna24-IT114-0>

End of Task 2

Task



Group: Reflection

Task #3: Add Screenshot of Wakatime

Weight: ~33%

Points: ~0.33

^ COLLAPSE ^

Details:

Note: The duration of time isn't directly related to the grade, the goal is to just make sure time is being tracked



Task Screenshots

Gallery Style: 2 Columns

4 2 1

Branches	
2 hrs 28 mins	M2-Java-Problems
Files	
1 hr 5 mins	M2/Problem1.java
52 mins	M2/Problem2.java
19 mins	M2/Problem3.java
8 mins	M2/.gitignore
3 mins	.git/COMMIT_EDITMSG
1 sec	Main.java

Showing the overall time

End of Task 3

End of Group: Reflection

Task Status: 3/3

End of Assignment

